

DRONE



GHOST IN TRANSLATION

MODE

DRONE

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MIX

SHAPE

GHOST IN
TRANSLATION

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1. INTRODUCTION

Drone is a multi-algorithm drone module based on my Motherboard platform.

Features

- 4 individual voices
- 16 algorithms
- 32 bit float DSP
- Stereo audio (48kHz, 32-bit)
- MIDI TRS Type A IN and OUT
- Auto calibration
- Either-way power

Technical Specifications

- **Module Width:** 8 HP
- **Depth:** 45 mm
- **Power Consumption:**
 - **+12V:** 75 mA
 - **-12V:** 3.5 mA
- **CV Input Range:** 0V to 5V

2. INSTALLATION

Turn off and disconnect your Eurorack case from AC mains power before installing this module. Connect the supplied power ribbon cable to the module and your busboard. This module is equipped with either-way power, so no magic smoke will be released if you plug it in backwards.

3. CONTROLS AND CV INPUTS

The front panel features controls for adjusting the tuning, timbral shape, and volume mix of the 4 voices.

All CV inputs read voltages between 0 and 5V.

Knobs

- **A B C D:** Sets the base pitch/tuning for each of the four individual voices.
- **MIX:** Turning this knob morphs the volumes of the four voices in a spiral motion.
- **SHAPE:** Adjusts the main character of the sound (such as morphing wavetables, filter sweeps, or FM intensity).

Each tuning knob has a different range, from 2 octaves to 5 octaves. This is to facilitate fine tuning rather than having a full 5 octaves on all voices.

CV Inputs

- **CV A, B, C, D:** 1V/Octave pitch CV inputs for each voice.
- **CV MIX:** Modulates the Mix parameter.
- **CV SHAPE:** Modulates the Shape parameter.
- **CV FM:** Linear frequency modulation for all four voices.

Navigation Button

- **Mode button:**
 - **Short Press:** Enters Algorithm Selection Mode (when in normal play mode) or cycles to the next algorithm (when in selection mode).
 - **Long Press (1 second):** Toggles Looper Mode (enters or exits).

4. MENU & NAVIGATION

Normal Play Mode

In normal play mode:

- The **8 LEDs at the top** display real-time levels of the control inputs and the main audio output.
- The **4 LEDs in the middle** show the volume of each of the four voices.

Algorithm Selection Mode

To change the active sound algorithm:

1. **Enter Menu:** Short press the **Mode button**. All 8 top LEDs will turn solid **Blue**, except for the active selection.
2. **Read Active Selection:** The active selection is shown by two blinking **Green** LEDs:
 - **Bank Indicator** (one of the first 4 LEDs)
 - **Algorithm Indicator** (one of the last 4 LEDs)
3. **Change Algorithm:** Tap the **Mode button** to cycle to the next algorithm.
4. **Save & Exit:** After **5 seconds** of inactivity, the module saves your choice and returns to play mode automatically.

5. THE 16 ALGORITHMS

The module contains 16 algorithms. Below is an overview of what each algorithm does and what the **SHAPE** knob controls.

Bank 1

#	Sound Description	What the SHAPE Control Does
1	Sine-Resonant Noise: A mix of warm, pure sine waves and soft resonant noise.	Blends from pure sine tones (0%) to wind-like resonant noise (100%).
2	Resonant Noise: Filters white noise into whistle-like pitched tones.	Sets Q-resonance. Turn up to morph from ocean-wave wind sounds to sharp, whistling pitches.
3	Sine-Phaser-Ring: Stereo phase-shifted sines combined with ring modulation.	Sets Ring Modulator speed (60Hz to 160Hz) to add metallic, sci-fi buzzes.
4	Sine-FM: Warm frequency modulation synthesis.	Sets modulator speed and modulation depth, introducing growling sidebands and metallic overtones.

Bank 2

#	Sound Description	What the SHAPE Control Does
1	Saw-Comparator: Gated, aggressive digital saw waves.	Sets the clipping threshold on the sum of the 4 saw waves.
2	Slink Filter: 8 parallel resonant bandpass filters swept by a complex LFO.	Adjusts sweep rate, ripple spacing, and filter density for vocal "formant" sweeps.
3	FM & Wavefolder: Aggressive, harmonically rich wavefolded FM voice.	Drives FM depth and folding intensity to add sharp digital bite and buzz.
4	Karplus-Strong Plucks: Simulated plucked and bowed string resonances.	Changes the brightness/filtering of the plucking excitation. Morphs from dark thuds to metallic plucks.

Bank 3

#	Sound Description	What the SHAPE Control Does
1	Gong & Bells: Inharmonic 4-operator FM chain recreating metallic bells.	Acts as modulation index. Morphs from simple sine tones to complex clangs and gong washes.
2	Dust-Triangle: Warm triangle drone mixed with crackling rain impulses.	Blends between crackling, pitched rain clicks (0%) and a clean, warm triangle wave drone (100%).
3	PWM (Pulse-Width Modulation): Thick, stereo-panned square waves with a custom foldback distortion waveshaper.	Morphs the pulse width (duty cycle) of the square waves to change the harmonic profile.
4	Resonant Saw & Saturation: Saturated sawtooth waves routed through a resonant lowpass filter.	Sweeps filter cutoff (200Hz to 6kHz), increases filter resonance, and drives the saturation level.

Bank 4

#	Sound Description	What the SHAPE Control Does
1	Wavetable 1: "Octopus": Complex 3D wavetable morphs with cascading modulation.	Scans smoothly through 128 waves to change the fundamental timbre.
2	Wavetable 2: "Bells": Metallic bell-like wavetable morphs with cascading modulation.	Scans smoothly through 128 waves to change the fundamental timbre.
3	Wavetable 3: "Fly By One": Vocal and sci-fi wavetable morphs with cascading modulation.	Scans smoothly through 128 waves to change the fundamental timbre.
4	Wavetable 4: "Yin Yang": Deep, harmonically complex wavetable morphs.	Scans smoothly through 128 waves to change the fundamental timbre.

6. BUILT-IN LOOPER

Drone features a built-in Looper to record and loop knob movements (TUNE A-D, MIX, SHAPE) up to **1 minute**, or **32 bars** when synchronized to a MIDI clock.

- **Enter / Exit Looper:** Hold the **Mode button** for **1 second**.
- **Start Recording:** Short press the **Mode button** while in **Idle** (all 8 LEDs solid **Red**).
- **Stop & Loop:** Short press the **Mode button** while recording (LEDs fill from **Red to Blue**). If MIDI clock is present, the loop stop is quantized to the next beat.
- **Reset / Clear:** Short press the **Mode button** during Playback to erase the loop and return to Idle.
- **Overdub:** Move any knob during Playback to record new values.
- **MIDI Sync:** recording and Playback automatically sync to external MIDI clock. A MIDI Start message resets Playback to the start of the loop.

7. MIDI IMPLEMENTATION

Drone listens to and transmits on **Channel 1**.

Control Change (CC) Mapping

- **CC 1:** Voice A Tune
- **CC 2:** Voice B Tune
- **CC 3:** Voice C Tune
- **CC 4:** Voice D Tune
- **CC 5:** Mix
- **CC 6:** Shape

Clock & Transport

- **MIDI Clock:** Synchronizes Looper recording and playback speed.
- **MIDI Start:** Resets loop playback to the beginning.

8. FIRMWARE UPDATE

The firmware update process is as follows:

1. **Power Off the Module:** Disconnect the module from power.
2. **Press and hold BOOTSEL:** Locate the physical **BOOTSEL** button on the microcontroller (Pico 2) on the back of the module, and hold it down.
3. **Connect to Computer:** While holding the **BOOTSEL** button, connect the Pico 2 to your computer via a USB cable.
4. **Release BOOTSEL:** Once the USB cable is plugged in, release the button. The Pico 2 will mount on your computer as a USB mass storage drive.
5. **Transfer the Firmware:** Locate the compiled firmware file `firmware/drone.ino.uf2`. Drag and drop (or copy) the `.uf2` file onto the drive.
6. **Automatic Reboot:** The Pico 2 will write the new firmware to its flash memory, automatically eject/unmount, and reboot with the updated firmware running.

9. TROUBLESHOOTING

No Sound Output

- Verify that the eurorack power ribbon cable is connected correctly.
- Sweep the **Mix** and **Shape** knobs to see if that makes any difference.
- Make sure your audio cable is patched securely.
- Switch off and switch on the module to restart the microcontroller.